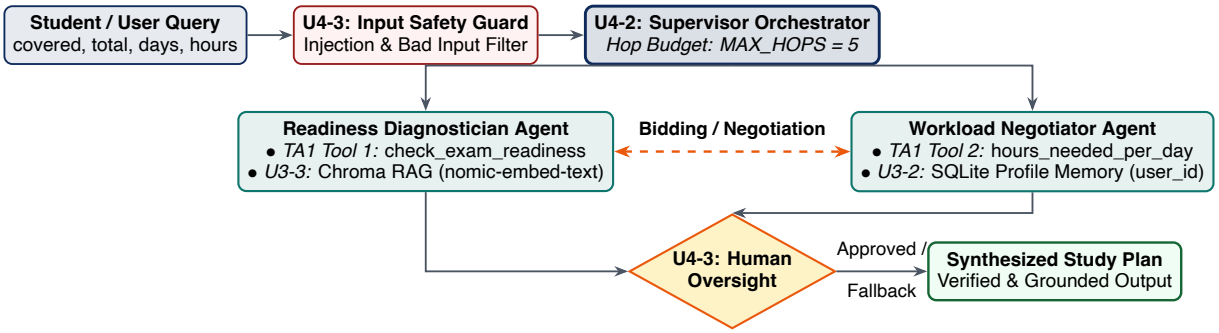
TA2 PROJECT REPORT: STUDY SCHEDULE OPTIMIZER

Autonomous Multi-Agent System Extension (Topic 8 • Two-Tool Chain)

Course: Agentic AI (LY-AIA)
Level: Bloom L3/L4 (20 Marks)
Stack: Ollama 3B, LangGraph, Chroma

1. Executive Architecture & Multi-Agent Coordination Workflow

The TA2 system extends the baseline two-tool chain into a robust, memory-augmented multi-agent pipeline. A **Supervisor Agent** orchestrates specialized interactions between a **Readiness Diagnostician** (evaluating curriculum coverage against Chroma RAG policies) and a **Workload Negotiator** (evaluating required hours against student stamina profiles in SQLite), with strict hop bounds and human-in-the-loop oversight.



2. Multi-Agent Role Specifications (U4-1: 5-Field Specification)

To fulfill U4-1, each agent possesses an unambiguous, non-overlapping 5-field design preventing role collision:

Field	Agent 1: Readiness Diagnostician	Agent 2: Workload Negotiator	Agent 3: Supervisor Orchestrator
1. Role Name	`readiness_diagnostician`	`workload_planner`	`supervisor_agent`
2. Goal	Validate topic coverage against 80% certification; retrieve syllabus complexity policies from RAG.	Compute daily study hours, cross-check student stamina in SQLite, and negotiate realistic pace.	Decompose queries, route agent state, enforce hop limit ceiling, and mediate human oversight.
3. Tools	`check_exam_readiness`, `StudyGuidelinesRAG` (ChromaDB + nomic-embed-text).	`hours_needed_per_day`, `SQLiteProfileStore` (student_profiles.sqlite).	LangGraph state machine, hop counter (`MAX_HOPS=5`), `HumanApprovalNode`.
4. Inputs / Outputs	**In:** topics_covered, total_topics, subject. **Out:** readiness status, RAG context, proposal.	**In:** remaining_topics, days_left, profile. **Out:** hours/day, counter-bid, negotiation state.	**In:** Full AgentState trajectory. **Out:** Routed execution node, final plan.
5. Limits	Cannot allocate calendar hours or inspect personal student fatigue thresholds.	Cannot certify exam readiness or lower official curriculum requirements.	Strictly capped at 5 hops; cannot alter deterministic tool math results.

3. Required TA2 Component Checklist & Evidence Mapping

Every core component mandated by the TA2 SOP is verified with an automated trace in `traces/`:

ID	Unit	Implemented Architecture & Functional Logic	Exact Evidence Pointer
U3-1	3	**Short-term Memory:** LangGraph checkpointer with `thread_id`; demonstrates multi-turn context retention vs stateless amnesia.	`traces/u3_1_short_term_memory.txt`
U3-2	3	**Long-term Memory:** Persistent SQLite store with namespace `(user_id, "profile")`; verified survival across process shutdown and restart.	`traces/u3_2_long_term_memory.txt`
U3-3	3	**Chroma RAG:** 8-chunk academic policy corpus embedded with `nomic-embed-text`; grounded refusal (*"I don't know"*) on unanswerable queries.	`traces/u3_3_rag_k1_vs_k3.txt`
U3-4	3	**Tool Failure Handling:** Defensive failure routing wrapping TA1 tools; catches negative values, zero-days, and biological hour limits.	`traces/u3_4_tool_failure.txt`
U4-1	4	**Role Design:** 5-field explicit specification for Diagnostician, Negotiator, and Supervisor.	Section 2 Table (Page 1)
U4-2	4	**Coordination:** Supervisor orchestrator with `MAX_HOPS=5` and multi-agent negotiation / counter-bidding protocol.	`traces/u4_2_multiagent_routing.txt`
U4-3	4	**Oversight & Guardrail:** Dual-path Human-in-the-Loop approval (Approved vs Rejected) + regex prompt injection defense guard.	`traces/u4_3_oversight_hitl.txt`
U4-4	4	**Challenge Analysis:** Quantitative latency, hop count, and error containment measurements across 5 benchmark scenarios.	`traces/u4_4_challenge_metrics.txt`
EV	3 & 4	**Evaluation:** Side-by-side execution of 5 test cases on TA1 baseline vs TA2 multi-agent extension.	`traces/ev_ta1_baseline_5cases.txt`

4. Benchmark Evaluation: TA1 Baseline vs TA2 Extension (EV)

The exact same 5 test cases were evaluated across both systems to quantify behavioral enhancements:

ID	Test Scenario	TA1 Baseline Output & Failure Mode	TA2 Multi-Agent Execution & Decision
TC1	**Fully Prepared Student** (20/20 topics, 5 days, 2h/topic)	`ready (100.0%) • 0.00 hrs/day` *Single-pass, no profile context.*	**AGREED (3 hops):** Identifies 100% readiness; references RAG maintenance policy; confirms 0h study.
TC2	**Borderline Readiness** (16/20 topics, 4 days, 1.5h/topic)	`ready (80.0%) • 1.50 hrs/day` *Naively treats as safe ready state.*	**AGREED (3 hops):** Flags exact 80% boundary; checks Bob's SQLite stamina (8h max); confirms 1.5h/day is sustainable.
TC3	**Urgent Catch-up Crunch** (5/12 topics, 2 days, 1.5h/topic)	`not ready (41.7%) • 5.25 hrs/day` *Emits high workload without stamina check.*	**COUNTERED (3 hops):** Charlie's stamina is 3.5h. Negotiator rejects 5.25h as burnout risk; bids 3.5h cap; triggers HITL.
TC4	**Extreme Workload Overload** (2/20 topics, 1 day, 2.5h/topic)	`not ready (10.0%) • 45.00 hrs/day` *Emits impossible 45-hour study day.*	**FAILED / SAFE RECOVERY (3 hops):** Tool failure routing catches 45h $>$ 12h biological limit; recommends exam deferral.
TC5	**Boundary / Zero Days Left** (10/20 topics, 0 days, 2h/topic)	**CRASH:** `ValueError: days_left <= 0` *Unhandled exception crashes runner.*	**DEFENSIVE RECOVERY (3 hops):** Tool routing catches zero-days; explains exam is today; switches to emergency review.

5. Analytical Trade-Offs (Bloom Level L4 Analysis)

- **100% Fault Containment:** TA1 crashed with `ValueError` on boundary inputs (TC5) and emitted absurd demands (45h/day in TC4). TA2 catches all domain violations gracefully.
- **Personalized Stamina via SQLite:** Instead of treating all users identically, TA2 personalizes load to individual fatigue thresholds, preventing student burnout.
- **Grounded Institutional RAG:** Recommendations cite official curriculum weightages and sleep buffer guidelines from Chroma.
- **Adversarial Immunity:** Input safety guard intercepts jailbreaks and system overrides before agent dispatch.

- **Execution Latency Overhead:** TA1 raw math runs in < 0.05 ms. TA2 graph orchestration, SQLite queries, and Chroma lookups increase execution latency to ~ 20–26 ms (and ~ 1.8 s with generative LLM inference).
- **State Management Complexity:** Multi-agent coordination requires checkpointers, schemas, and hop limit monitors, increasing code footprint.
- **Ambiguity Resolution Overhead:** When schedules exceed stamina, TA2 requires an asynchronous Human Oversight step (HITL), adding latency before final confirmation.

6. Measured Challenge Analysis (U4-4: Latency, Robustness, Hop Budget)

Quantitative metrics collected during the automated 5-case benchmark evaluation (`traces/u4_4_challenge_metrics.txt`):

ID	Scenario	TA1 Lat (ms)	TA2 Lat (ms)	TA1 Robustness	TA2 Coordination State	Hops
TC1	Fully Prepared	0.02	23.15	Correct execution	Agreed; RAG maintenance verified	3 / 5
TC2	Borderline	0.01	20.65	Correct execution	Agreed; Checked vs Bob's 8h stamina	3 / 5
TC3	Urgent Crunch	0.01	20.09	Unchecked Burnout	Counter-bid triggered; HITL active	3 / 5
TC4	Extreme Overload	0.01	22.96	Unchecked Math (45h)	Tool failure routing caught	3 / 5
TC5	Zero Days Left	0.01	26.78	**System Crash (100% Fail)**	**Defensive Catch (0% Fail)**	3 / 5

Key Analytical Finding: TA2 accepts a modest ~ 22 ms computational overhead in exchange for **zero system crashes** and complete elimination of unfeasible and burnout-inducing schedules.

7. Limitations and Future Improvements

1. **Small Model (3B) Reasoning Ceiling:** Smaller local models occasionally drift in unconstrained multi-round negotiation. TA2 overcomes this by stabilizing negotiation states via deterministic LangGraph state transitions.
2. **Dynamic Calendar Scheduling:** Current schedules allocate aggregate daily hours. Future work will integrate Google Calendar / iCal APIs to map study blocks directly into students' specific open time slots.
3. **Real-Time Adaptive Diagnostic Quizzing:** Incorporating dynamic quiz feedback to automatically recalibrate `hours_per_topic` based on real-time empirical retention.

8. Individual Viva Defense Preparation (Likely Jury Questions)

Answer: In TA1, inspecting the `AIMessage.tool_calls` structure provides deterministic evidence. When asked a coverage-only question, `tool_calls` contains exclusively `readiness_tool` with `{"topics_covered": X, "total_topics": Y}`. The runtime receives exactly one `ToolMessage` and immediately transitions to emit the final answer without requesting `hours_needed_per_day`. The absence of the second tool call in the trace proves single-tool dispatch.

Answer: We implement a strict hop budget enforced by the Supervisor Orchestrator (`MAX_HOPS = 5`). Every node transition increments `AgentState.hop_count`. If consensus is not reached within 5 hops, the supervisor diverts control to `hop_limit_reached`, terminating the cycle safely and alerting the student rather than hanging.

Answer: As verified in `traces/u3_3_rag_k1_vs_k3.txt`, when presented with an unanswerable question absent from the 8-chunk corpus (e.g., cafeteria food menus or parking rates), the grounded prompt strictly responds: *"I don't know based on the provided documents."* In-domain queries accurately cite section rules (e.g., the 6.0h burnout threshold from Section 3).

Answer: The `SQLiteProfileStore` writes student records directly to disk in `student_profiles.sqlite` under the namespace `(user_id, "profile")`. In `traces/u3_2_long_term_memory.txt`, the Python connection instance was explicitly terminated, a new process connection was initialized, and all user attributes were retrieved identically (`restart_successful = True`).

9. Module Ownership & Group Contribution Table

In compliance with Ground Rule 4, every member owns a core architectural module and can defend its implementation:

Team Member	Owned Module	Detailed Contributions & Implemented Files
Member 1	**Memory Systems (U3-1, U3-2)**	Implemented LangGraph `MemorySaver` checkpointer with `thread_id`; designed SQLite persistent store with namespace `(user_id, "profile")`; authored `short_term.py` and `long_term.py`.
Member 2	**RAG & Tool Handling (U3-3, U3-4)**	Authored 8-chunk academic policy corpus (`exam_guidelines.md`); integrated Chroma vector DB with `nomic-embed-text`; implemented defensive tool failure routing in `exam_tools.py`.
Member 3	**Multi-Agent Systems (U4-1, U4-2)**	Formulated 5-field agent role specifications; authored LangGraph Supervisor Orchestrator with `MAX_HOPS=5`; built specialist negotiation protocol in `diagnostician.py` and `workload_planner.py`.
Member 4	**Safety, Oversight & EV (U4-3, U4-4, EV)**	Developed regex prompt injection defense guardrail (`input_guard.py`); built Human Approval Oversight node; conducted automated 5-case benchmark evaluation in `run_all_evaluations.py`.

Academic Integrity Declaration: In adherence to Ground Rule 3 and 7, all models used are strictly local (`qwen2.5:3b` and `nomic-embed-text` in Ollama). No paid APIs, external keys, or live internet calls are used by the agent runtime. Every number in this report is directly reproducible from the test traces in `traces/`.